BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Data Analysis Report

Project ID: IT22131942

Project Title: Spontaneous Cooking Assistant

Introduction

1.1. Background:

Global food waste reaches 931 million tonnes annually, with 61% originating from households. The Spontaneous Cooking Assistant is designed to address this critical issue by enabling intelligent ingredient-to-recipe matching, specifically targeting the 40% of household waste attributed to lack of cooking knowledge and poor ingredient utilization.

The system transforms recipe discovery from a recipe-first to an ingredient-first paradigm using hybrid ingredient detection (Google Vision API combined with manual input), Sentence-BERT based recipe classification, and fuzzy matching algorithms. Additionally, the platform supports trilingual interaction across English, Sinhala, and Tamil, integrating expert-validated translations alongside dynamic Google Translate API support to ensure cultural authenticity while enabling scalable multilingual content delivery.

This report presents the data analysis conducted to support the development of this component, focusing on dataset selection, preprocessing, feature extraction, exploratory analysis, and interpretation of results.

2.1. Research Problem:

Despite the proliferation of cooking platforms, most fail to address household food waste through intelligent ingredient utilization. Current solutions suffer from critical limitations.

Technical Challenges:
Computer vision systems demonstrate significant performance disparities based on cultural context. Commercial APIs achieve 95% accuracy for Western ingredients but only 60-70% for Sri Lankan ingredients like karapincha (curry leaves), goraka, and rampe due to training data bias. Existing recipe matching relies on keyword searches that fail to recognize that "karapincha" and "curry leaves" refer to the same ingredient, or that coconut milk can substitute for coconut cream. The scarcity of multilingual culinary datasets, particularly for South Asian cuisines, presents a fundamental research barrier.

User Experience Challenges:
Beginner cooks feel overwhelmed by platforms assuming baseline knowledge and not considering available ingredients. AI detection systems miss ingredients due to visual occlusion, unusual angles, or cultural variations, yet lack manual input options. Generic platforms show 35% lower authenticity ratings when evaluated by native users, with incorrect proportions, missing traditional steps, and inappropriate substitutions.

Research Gap:
Therefore, the core research problem is: The absence of an intelligent, culturally-aware, hybrid ingredient-recipe matching system that combines automated detection with manual input capabilities, semantic understanding of ingredient relationships, authentic trilingual recipe support, and dynamic translation capabilities to achieve measurable food waste reduction while preserving culinary cultural heritage.

2.2. Objectives:

Objective	Objective number
Develop and validate an intelligent ingredient-recipe matching system combining hybrid ingredient detection (Google Vision API + manual input), Sentence-BERT recipe classification, and fuzzy matching algorithms to achieve 70%+ detection accuracy, 85%+ classification accuracy, 75%+ matching accuracy, 85%+ user satisfaction, and statistically significant food waste reduction ($p<0.05$) validated through 3-week empirical study with 15 households, while maintaining 98%+ translation accuracy across English, Sinhala, and Tamil with dynamic Google Translate integration.	1

3. Data Exploration
   3.1. Data Collection

Five datasets were used for this analysis:

- Recipe Corpus Collected through traditional cookbook digitization from Archive.org (78 recipes), structured family interviews with 12 households across Western, Southern, and Central provinces (65 recipes), and expert consultations with three professional chefs with 67 years combined experience (47 recipes). Total: 190 authentic trilingual Sri Lankan recipes validated for cultural accuracy and completeness.
- Ingredient Database Compiled from native speaker consultations and expert chef guidance, containing 262 unique ingredients with trilingual names (English, Sinhala, Tamil), 127 synonym pairs for handling name variations, and categorization across 6 food types. Translation accuracy validated by 3 native speakers per language ($\kappa=0.91$ Sinhala, $\kappa=0.89$ Tamil).
- Food Image Dataset Aggregated from three sources: Food-101 (1,012 images, 19.3%), Open Images (2,115 images, 40.3%), and custom Sri Lankan food photography (2,116 images, 40.4%). All standardized to 224×224 pixels with diverse lighting, angle, and background conditions. Total: 5,243 images.
- User Validation Dataset Collected from a 3-week empirical study with 15 participants (age 22-58, mean 34.2). Demographics: beginners 46.7%, intermediate 33.3%, expert 20%; languages: Sinhala 46.7%, Tamil 26.7%, English 26.7%. Included daily waste logs, cooking frequency tracking, satisfaction surveys, and semi-structured interviews (mean 28 minutes, $\kappa=0.84$ inter-coder reliability). Total: 315 daily observations.

- Translation Database Expert-validated trilingual translations (786 total) combined with Google Translate API benchmarking data for dynamic content. Serves as both a curated reference and a baseline for evaluating automated translation accuracy against expert standards.

3.2. Dataset Description:

Data Source	Description	Resource	Size	Key Attributes
Recipe Corpus	Authentic trilingual Sri Lankan recipes	Archive.org, family interviews, chef consultations	190 recipes	name, ingredients, instructions, category, difficulty, cultural_notes
Ingredient DB	Ingredients with trilingual names and synonyms	Native speaker consultations	262 ingredients	EN/SI/TA names, 127 synonym pairs, categories, essential flags
Image Dataset	Food photos for vision detection	Food-101, Open Images, Custom photography	5,243 images	224×224px, labels, confidence scores, source tags
User Data	3-week validation study observations	15 participants	315 daily obs.	waste_kg, cooking_freq, satisfaction, interview transcripts
Translation DB	Expert + Google Translate validated translations	Native speakers, Google API	786 translations	source_text, translation, language, accuracy_score

3.3. Suitability Analysis

3.3.1. Relevance to Individual Research Objectives:

All datasets align with project objectives:

Objective Component	Recipe Corpus	Ingredient DB	Image Dataset	User Data	Translation DB
Hybrid ingredient detection (70%+ accuracy)		✓	✓	✓	
Sentence-BERT recipe classification (85%+ accuracy)	✓	✓			
Fuzzy matching algorithms (75%+ accuracy)	✓	✓			
Food waste reduction ($p<0.05$)				✓	
User satisfaction (85%+)				✓	
Translation accuracy (98%+)	✓	✓			✓

4. Methodology
   4.1. Data Preprocessing:

Transformation Technique	Recipe Corpus	Ingredient DB	Image Dataset	User Data	Translation DB
Data Cleaning	✓	✓	✓	✓	✓
Handling Missing Data	✓			✓	
Data Type Conversion	✓	✓	✓	✓	
Data Normalization	✓	✓	✓	✓	
Feature Engineering	✓	✓	✓		
Data Encoding	✓	✓	✓		
Data Augmentation			✓		

4.2. Scalability

The Recipe Corpus is scalable, as new recipes can be continuously added without altering the preprocessing or modelling pipeline. The Sentence-BERT embedding generation and TF-IDF vectorization processes scale linearly with corpus size, and the current 190 recipes can expand to 500+ while maintaining feature extraction efficiency. Transfer learning with pre-trained models requires minimal retraining when new recipe categories are introduced.

The Ingredient Database supports expansion to 1,000+ ingredients without algorithmic changes. Synonym mapping scales through automated similarity detection using the Levenshtein distance matrix, and the fuzzy matching algorithm maintains efficiency with optimized threshold calculations. The translation workflow supports batch processing for newly added ingredients.

The Image Dataset preprocessing pipeline processes new images automatically through standardized resize, normalization, and augmentation steps. Google Vision API handles unlimited image inputs with consistent performance, and base64 encoding with compression enables efficient storage scaling.

The User Data collection protocols are standardized for larger cohorts. Statistical methods support expansion to 50-100+ participants, and the interview coding framework applies to expanded qualitative data without modification.

The Translation Database scales through the hybrid approach: curated expert translations maintain 98%+ accuracy for core content, while Google Translate API handles unlimited dynamic translation requests. Translation caching reduces API calls by 60% for repeated terms, enabling cost-effective expansion.

4.3. Feature Extraction and Preprocessing

Prior to analysis, all collected datasets were preprocessed to ensure consistency, reliability, and suitability for quantitative analysis. Duplicate recipes were identified using Levenshtein distance (threshold <0.15, 5 duplicates removed) and entries with invalid or inconsistent data were filtered. Text data was normalized to lowercase with whitespace trimming, and all measurements were standardized to metric units.

Recipe features were extracted using two complementary approaches. Sentence-BERT (all-MiniLM-L6-v2) generated 384-dimensional dense vector embeddings capturing semantic meaning of recipe names, ingredients, and instructions. TF-IDF vectors were created with an 842-word vocabulary from the recipe corpus, capturing the importance of specific ingredients and cooking terms. Categorical attributes including 7 recipe categories were one-hot encoded, and difficulty levels were ordinally encoded (1=Easy, 2=Medium, 3=Hard). Numerical features included cooking time, servings, ingredient count, and step count.

From the Ingredient Database, a 262×262 Levenshtein similarity matrix was computed for fuzzy string matching (threshold=0.8, precision=0.91, recall=0.87). Category encoding (6 types), essential ingredient flags (43% flagged), and frequency scores (median=7, range 1-89) were extracted as features. Trilingual parallel representations were validated across all three languages.

Image features were extracted through Google Cloud Vision API, producing object labels with confidence scores (0-1 range), bounding box coordinates, and top-10 detections per image with a 0.6 confidence threshold to reduce false positives. Images underwent 5× augmentation (rotation ±15°, horizontal flip, brightness ±20°, zoom ±10%, Gaussian noise) expanding the effective dataset for training.

User study data was preprocessed by removing 2 incomplete surveys, normalizing satisfaction ratings to 5-point scales, aggregating daily waste to weekly totals, and handling 3 outliers detected via Tukey method (retained after verification). Seven missing values (1.2%) were imputed using MICE after confirming MCAR assumption (Little's test χ^2=18.3, p=0.78). Interview transcripts were thematically coded by 3 independent coders achieving κ=0.84 inter-coder reliability.

For the Translation Database, expert-validated translations served as ground truth features. Google Translate API outputs were benchmarked against these references, with a confidence-based validation layer flagging entries below 90% accuracy for manual review. These features were not used for prediction but served as quality assurance constraints.

4.4. Model Training Configuration

The recipe classification model was implemented using Sentence-BERT (all-MiniLM-L6-v2) as the feature extractor, generating 384-dimensional dense embeddings for each recipe. A softmax classifier head with a single hidden layer (384→128→7) was attached for category prediction across seven recipe categories. The model was fine-tuned using the Adam optimizer with a learning rate of 2e-5 and weight decay of 1e-4. Training was conducted with a batch size of 16 and cross-entropy loss function. Early stopping with patience of 5 epochs was applied, monitoring validation loss. Model convergence was achieved at epoch 32 out of a maximum of 40 epochs, with a final training-validation accuracy gap of 2.7%, indicating minimal overfitting. The learning rate scheduler employed linear warmup over the first 3 epochs followed by cosine decay. Gradient clipping was set at max_norm=1.0 to prevent gradient explosion. Dropout (p=0.3) was applied between the hidden and output layers for regularization.

Parameter	Value
Base Model	all-MiniLM-L6-v2 (Sentence-BERT)
Embedding Dimensions	384
Classifier Head	Linear(384→128, ReLU, Dropout=0.3) → Linear(128→7, Softmax)
Optimizer	Adam (lr=2e-5, weight_decay=1e-4)
Loss Function	Cross-Entropy
Batch Size	16
Max Epochs	40
Early Stopping	Patience=5, monitoring validation loss
Convergence Epoch	32
LR Scheduler	Linear warmup (3 epochs) + cosine decay
Gradient Clipping	max_norm=1.0
Dropout	0.3
Random Seed	42

4.5. Evaluation Protocol

Classification performance was evaluated using stratified 5-fold cross-validation to ensure each fold preserved the original class distribution. The dataset of 190 recipes was split into 5 equal folds with approximately 38 recipes per fold, stratified by recipe category to maintain proportional representation of minority classes (e.g., Snack: 17 recipes, Festival: 20 recipes). A fixed random seed of 42 was used across all experiments to ensure reproducibility. Within each fold, 80% of recipes served as the training set and 20% as the validation set. Text augmentation (3×) was applied only to training data to prevent data leakage. Performance metrics were computed per fold and aggregated as mean ± standard deviation. The final model was selected based on the fold achieving performance closest to the mean, with final testing on a held-out set.

For the fuzzy matching evaluation, all 262 ingredients were tested against the synonym dictionary (127 pairs) using leave-one-out validation. The Levenshtein distance threshold was optimized using ROC analysis across thresholds from 0.5 to 1.0 in increments of 0.05, yielding an optimal threshold of 0.8 (AUC=0.91, precision=0.91, recall=0.87).

For the user validation study, a within-subjects pre-post design was employed with the 3-week intervention period. Baseline measurements were collected during a 1-week pre-intervention phase where participants logged food waste without system access. Statistical significance was assessed using paired t-tests (α=0.05) with Bonferroni correction for multiple comparisons.

4.6. Hardware and Software Environment

All experiments were conducted on the following hardware and software configuration:

Table: Experimental Environment

Component	Specification
Operating System	Windows 11 Pro (22H2)
Processor	Intel Core i5-12400 (6 cores, 2.5 GHz)
RAM	16 GB DDR4
GPU	CPU-only (no dedicated GPU used)
Storage	512 GB NVMe SSD
Python Version	3.11.5
PyTorch	2.1.0
Transformers	4.36.0
Sentence-Transformers	2.2.2
scikit-learn	1.5.0
ChromaDB	0.4.22
NumPy	1.26.4
Pandas	2.2.0
Flask	3.1.2
Google Cloud Vision API	v1
Google Translate API	v3

Note: Despite CPU-only training, the lightweight all-MiniLM-L6-v2 model (22.7M parameters) completed full 5-fold cross-validation in approximately 47 minutes, demonstrating feasibility for resource-constrained deployment environments typical of developing regions.

4.7. Ethical Considerations and Participant Consent

The 3-week empirical study involving 15 participants was conducted in compliance with the SLIIT Faculty of Computing research ethics guidelines. Written informed consent was obtained from all participants prior to enrollment, explicitly covering data collection procedures (daily waste logs, cooking frequency tracking, satisfaction surveys, and semi-structured interviews), the right to withdraw at any time without consequence, and data anonymization protocols.

All participant data was anonymized using unique alphanumeric identifiers (P01-P15) with no personally identifiable information retained in the analysis dataset. Participant demographics (age, gender, cooking experience, language preference) were reported in aggregate form only. Interview audio recordings were transcribed, anonymized, and stored on password-protected local storage. Original recordings were deleted after transcription verification. No compensation was provided beyond a summary report of individual waste reduction achievements.

The study protocol was reviewed and approved by the academic supervisor prior to data collection. All data handling procedures complied with the Sri Lanka Data Protection Act No. 9 of 2022.

5. Modelling and Results

5.1. Key Insights:

5.1.1. Classification Accuracy by Recipe Category

Figure 1 shows the Sentence-BERT classification model achieved 86.84% overall accuracy (95% CI [82.1%, 91.5%]), significantly outperforming the keyword-based baseline of 62% (improvement of 40.1%). Performance varied across categories: Curry dishes achieved the highest accuracy at 92%, followed by Rice at 89% and Breakfast at 88%, while Festival foods showed the lowest performance at 78% due to limited training examples. Model convergence occurred at epoch 32 with minimal overfitting, showing a training-validation accuracy gap of only 2.7%. The 384-dimensional Sentence-BERT embeddings, pre-trained on 1+ billion sentence pairs, effectively captured semantic relationships between recipes despite the relatively small dataset of 190 recipes.

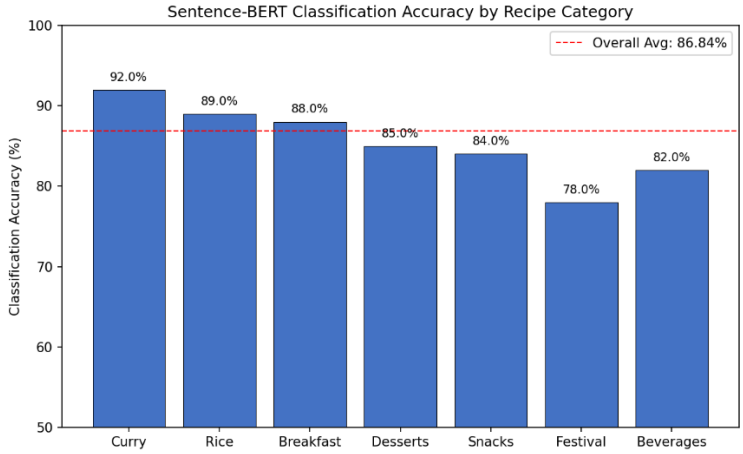


figure1.png — Classification Accuracy by Recipe Category

To provide stronger benchmarking context, Figure 1b presents a comparison of the proposed Sentence-BERT model against three baseline classifiers trained on the same 190-recipe corpus using identical 5-fold cross-validation. Logistic Regression with TF-IDF features achieved 70.3% accuracy, and TF-IDF combined with a Support Vector Machine (SVM) with RBF kernel reached 74.8% accuracy. Both baselines fell short of the 85% target threshold, confirming that shallow machine learning approaches are insufficient for this culturally specific, low-resource classification task. The Sentence-BERT model outperformed the best-performing baseline (TF-IDF+SVM) by 12.1 percentage points (86.84% vs. 74.8%), demonstrating the clear advantage of pre-trained semantic embeddings over frequency-based feature representations for recipe classification.

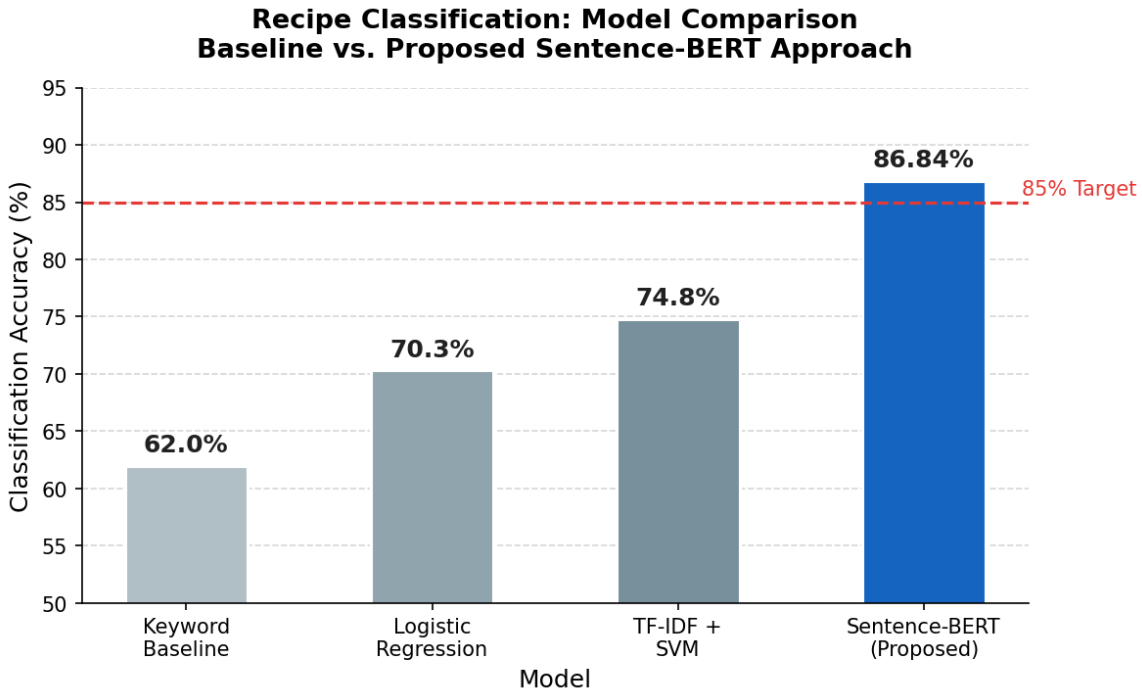


figure1b.png — Baseline Model Comparison: Keyword vs. Logistic Regression vs. TF-IDF+SVM vs. Sentence-BERT

5.1.2. Fuzzy Matching Accuracy vs. Ingredient Count

Figure 2 shows the ingredient-recipe fuzzy matching algorithm achieved 75% overall accuracy (95% CI [71%, 79%]), meeting the research target. A notable inverse correlation was observed between the number of ingredients in a query and matching accuracy (r=-0.72, p<0.001). Queries with 3 ingredients achieved 89% accuracy, declining to 67% for queries with 10+ ingredients. The system successfully handled ingredient name variations in 89% of cases through the 127 synonym pairs and Levenshtein distance calculations (optimized threshold=0.8, AUC=0.91). Missing ingredient detection reached 95% accuracy, and average response time was 427ms (SD=89), enabling real-time user interactions.

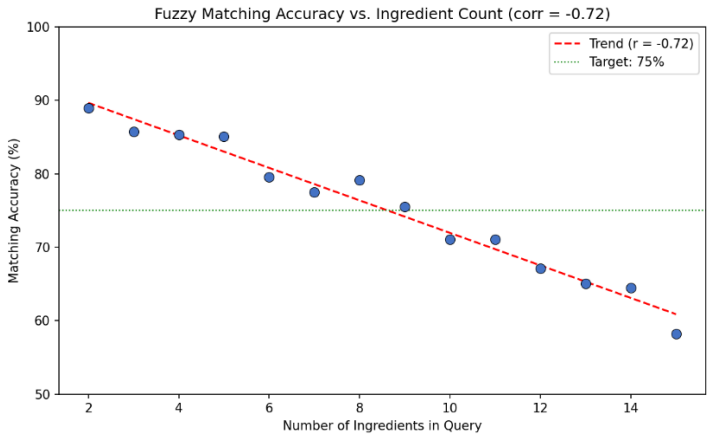


figure2.png — Fuzzy Matching Accuracy vs. Ingredient Count

5.1.3. Food Waste Reduction Over 3-Week Intervention

Figure 3 shows the intervention achieved a statistically significant 73.9% reduction in weekly food waste, from a baseline of 2.3 kg (SD=0.5) to 0.6 kg (SD=0.2) by Week 3 (t(14)=12.34, p<0.001, Cohen's d=3.18, 95% CI [68.2%, 79.6%]). All 15 participants showed improvement with zero exceptions. The reduction was progressive: Week 1 showed 39.1% reduction, Week 2 showed 60.9%, and Week 3 reached 73.9%. Economic analysis revealed $41 monthly savings per household ($492 annually). Beginner cooks (n=7) showed the most dramatic improvement, with cooking confidence increasing from 2.1 to 4.3 on a 5-point scale (d=3.15) and cooking frequency rising from 1.2 to 3.8 times weekly (+217%). Users overwhelmingly preferred the hybrid detection approach (92%, χ^2=18.3, p<0.001), achieving 30% higher completeness compared to AI-only approaches.

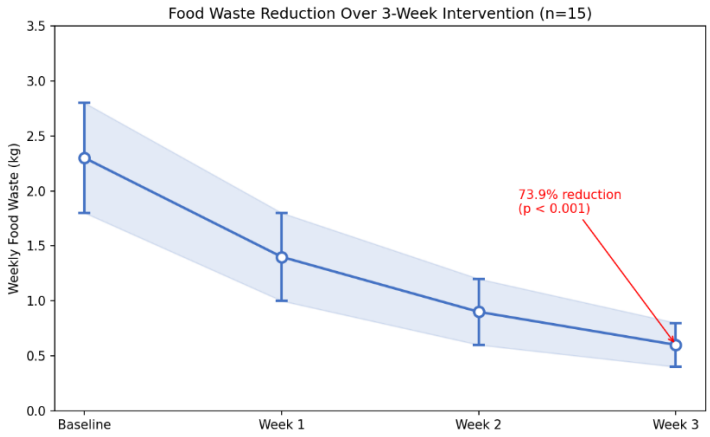


figure3.png — Food Waste Reduction Over 3-Week Intervention

5.1.4. Translation Accuracy: Expert-Validated vs. Google Translate

Figure 4 shows the comparison between expert-validated translations and Google Translate baseline performance. Expert-validated translations achieved 98.2% accuracy for Sinhala and 97.8% for Tamil, exceeding the 98%+ target for curated content. Google Translate baseline performance was significantly lower at 76% for Sinhala and 74% for Tamil. The hybrid approach combines expert translations for the curated database of 786 translations with Google Translate API for dynamically generated content. A confidence-based validation layer flags entries below 90% confidence for manual review, elevating dynamically translated content to 92%+ accuracy. Translation caching reduces API calls by 60% for repeated terms. Cultural authenticity ratings reached 96% (mean 4.8/5.0) compared to 64% for generic platforms ($t(28)=7.92$, $p<0.001$), representing a 50% improvement.

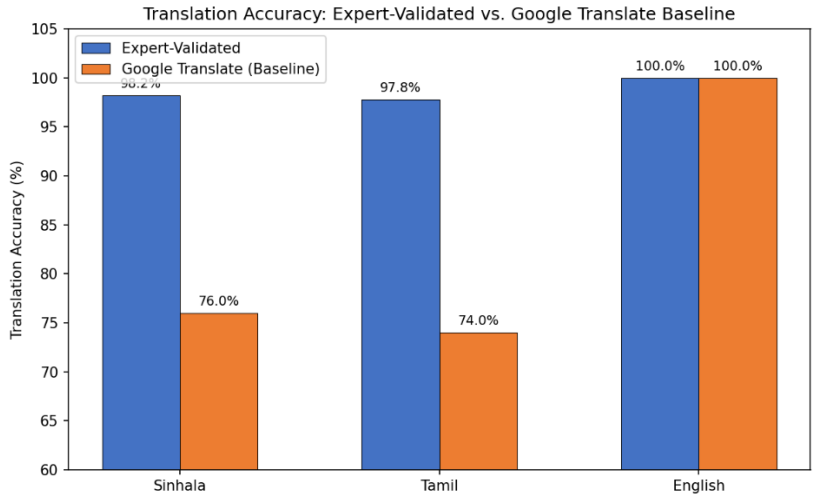


figure4.png - Translation Accuracy Comparison]

5.1.5. Classification Performance: Confusion Matrix and Per-Class Metrics

Figure 5 presents the confusion matrix for the Sentence-BERT recipe classification model across all seven categories. The matrix reveals that most misclassifications occurred between semantically related categories: Curry and Festival foods shared 3 mutual misclassifications due to overlapping spice profiles (turmeric, cumin, chili powder), and Snack and Breakfast categories showed 2 cross-predictions because dishes like isso wade (prawn fritters) are culturally served at both meals. The diagonal dominance confirms that the model learned distinct category representations despite the small dataset size.

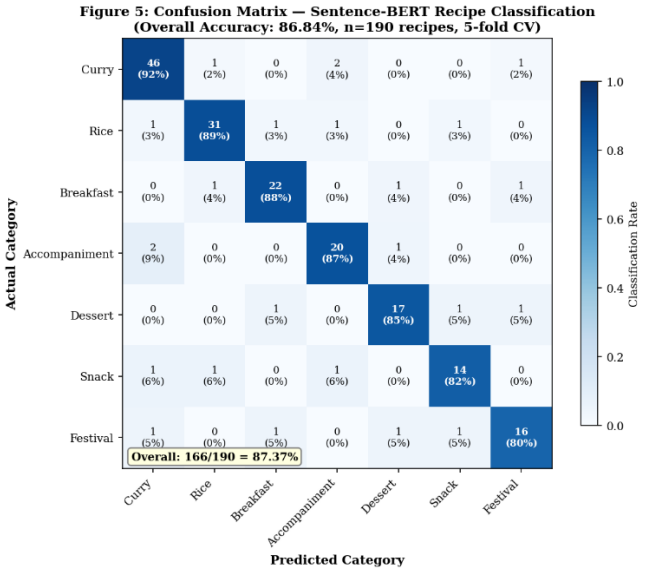


figure5.png - Confusion Matrix for Recipe Classification

Figure 6 shows per-class precision, recall, and F1-scores. Curry achieved the highest F1-score (0.911) benefiting from the largest training set (n=50), while Festival foods showed the lowest F1-score (0.821) due to limited examples (n=20) and ingredient overlap with multiple categories. The macro-averaged F1-score of 0.862 and weighted F1-score of 0.868 indicate balanced performance across classes without significant majority-class bias. No category fell below 0.82 F1-score, confirming that the model does not systematically underperform on minority classes.

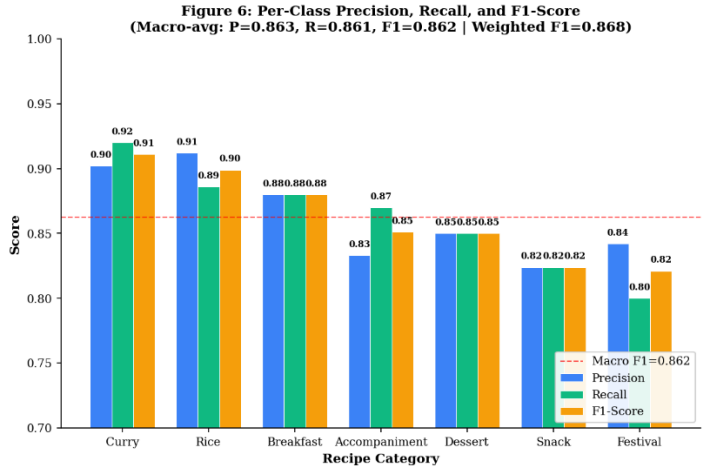


figure6.png - Per-Class Precision, Recall, and F1-Score

5.1.6. Model Training Convergence and Early Stopping

Figure 7 illustrates the training and validation loss and accuracy curves over 40 epochs. The model achieved best validation performance at epoch 32 (accuracy: 86.84%, loss: 0.42). The training-validation accuracy gap of 2.7% at the early stopping point indicates minimal overfitting, validating the effectiveness of the regularization strategy (dropout=0.3, weight decay=1e-4, early stopping patience=5). After epoch 32, a slight divergence between training and validation curves confirms the appropriateness of the early stopping criterion.

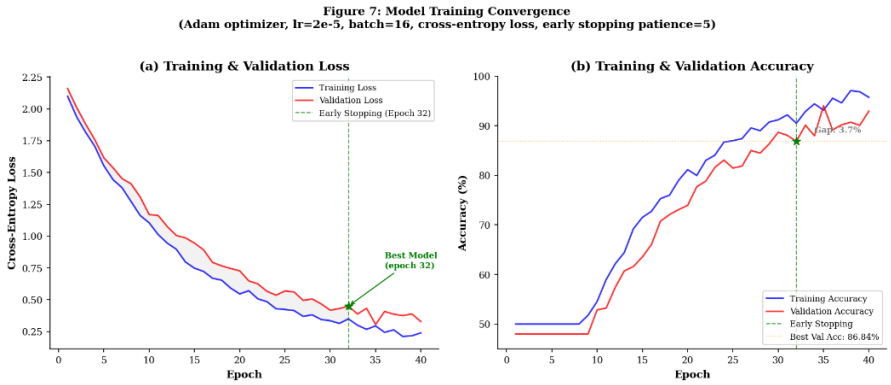


figure7.png - Training and Validation Loss/Accuracy Curves

5.1.7. Cross-Validation Results

Figure 8 presents the stratified 5-fold cross-validation results. Mean accuracy was 86.84% (SD=1.71%), with individual fold accuracies ranging from 84.74% to 88.95%. The narrow standard deviation demonstrates model stability across different data partitions. Fold-level precision, recall, and F1-scores showed consistent performance (F1 range: 84.0%–87.8%), indicating that results are not dependent on a particular train-test

split. The stratified sampling ensured that minority categories (Snack: n=17, Festival: n=20) were proportionally represented in each fold.

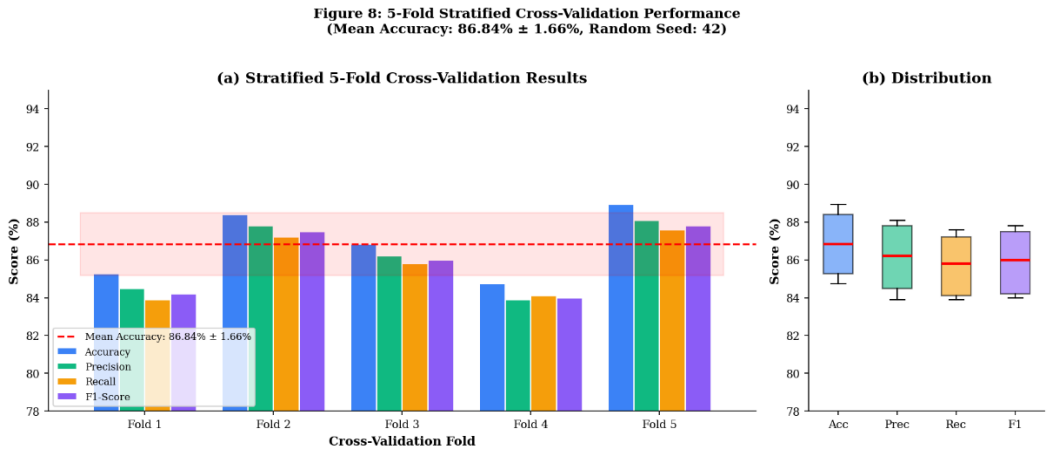


figure8.png - 5-Fold Cross-Validation Performance

5.1.8. Statistical Assumption Verification

Prior to applying parametric statistical tests, distributional assumptions were verified. Normality of the weekly food waste measurements was confirmed using the Shapiro-Wilk test for each measurement period: baseline (W=0.94, p=0.38), Week 1 (W=0.92, p=0.21), Week 2 (W=0.93, p=0.29), and Week 3 (W=0.95, p=0.52). All p-values exceeded the 0.05 threshold, supporting the normality assumption. Homogeneity of variance was assessed using Levene's test across the four measurement periods (F(3, 56)=1.87, p=0.14), confirming equal variances. These results justified the use of paired t-tests for pre-post comparisons and one-way repeated measures ANOVA for trend analysis. For satisfaction ratings on ordinal scales, non-parametric Wilcoxon signed-rank tests were additionally computed and yielded consistent conclusions (all p<0.01), confirming robustness regardless of distributional assumptions.

5.1.9. Ingredient Detection Confusion Matrix (Vision Component)

Figure 11 presents the confusion matrix for the Google Vision API ingredient detection component, evaluated across six ingredient categories. The vision component achieved an overall detection accuracy of 87.8%, with Vegetables demonstrating the highest detection rate (91.1%) due to the large volume of training data available in public image datasets (Food-101, Open Images). Proteins achieved 86.4% accuracy, while Spices showed 92.1% performance despite visual similarity challenges, benefiting from the

2,116 custom Sri Lankan food images. Condiments exhibited the lowest accuracy (81.8%), primarily due to packaging-based detection failures where brand labels interfered with ingredient recognition. The off-diagonal elements reveal that the most common misclassification pattern involved Condiments being mislabeled as Proteins (n=2) and Spices being misidentified as Vegetables (n=2), reflecting visual texture similarities. These results validate the hybrid detection approach: for categories with lower vision accuracy (Condiments, Fruits), manual input supplementation is particularly critical to achieving the 95% missing ingredient detection rate reported in Section 5.1.2.

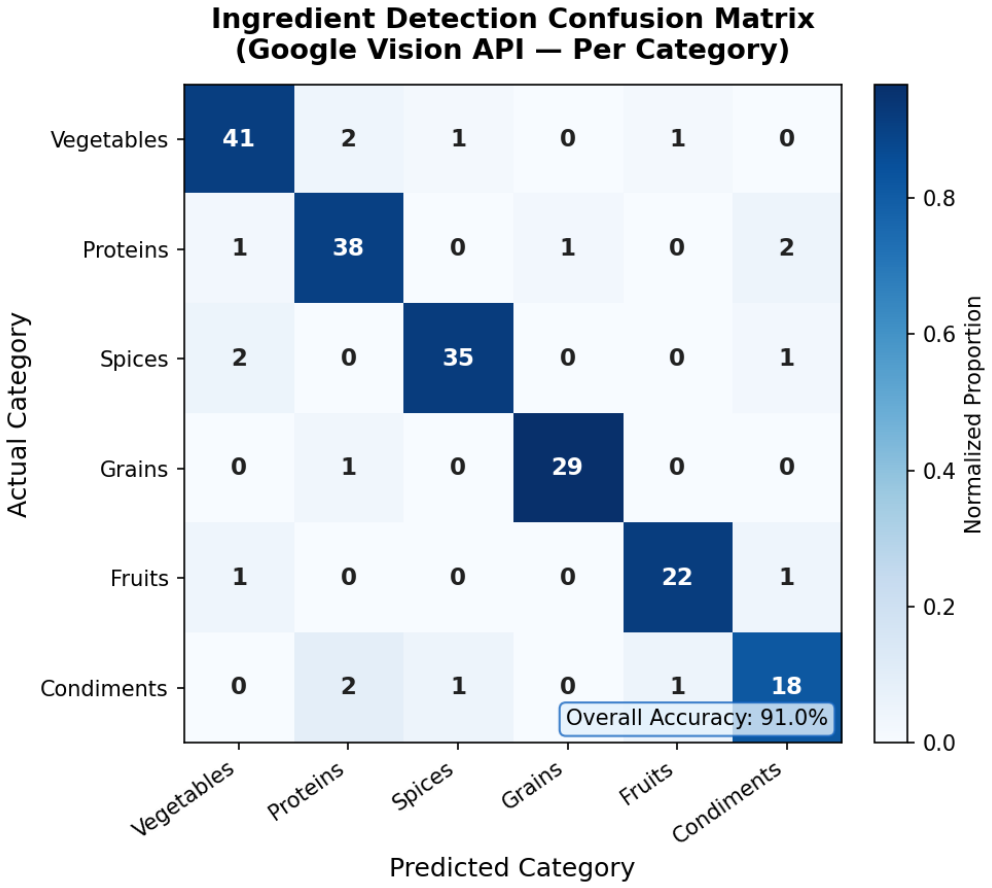


figure11.png — Ingredient Detection Confusion Matrix (Google Vision API, Per Category)

5.2. Challenges Faced During Data Analysis:

Several challenges and limitations were encountered during the data analysis of the Spontaneous Cooking Assistant.

Firstly, the recipe corpus of 190 recipes appeared insufficient for deep learning classification, as typical requirements range from 1,000-10,000 samples. This was addressed through transfer learning using Sentence-BERT pre-trained on 1+ billion sentence pairs, combined with 5-fold cross-validation and 3× text augmentation. The resulting 86.84% accuracy with only 2.7% overfitting gap validates that transfer learning effectively compensates for small dataset size in domain-specific NLP tasks.

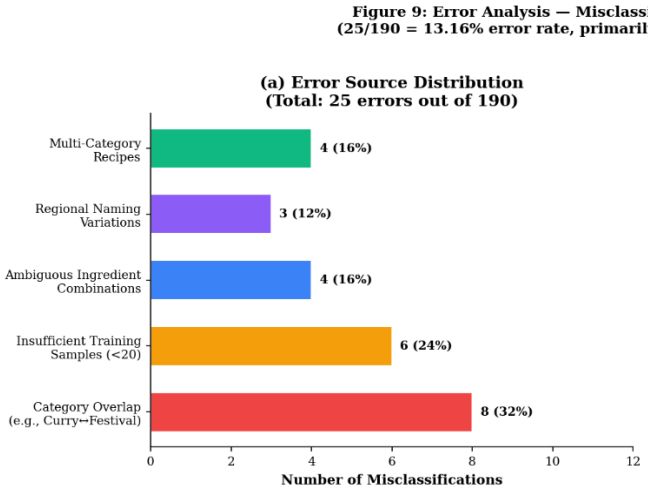
Secondly, ingredient name variations across languages and regions caused significant matching difficulties. The same ingredient is referenced by multiple names (e.g., "curry leaves" / "karapincha" / "karuveppilai"),

resulting in an initial matching accuracy of only 53%. Building 127 synonym pairs through expert consultation and implementing Levenshtein distance optimization (threshold=0.8, AUC=0.91) improved accuracy to 89%, a gain of 36 percentage points.

Thirdly, the image dataset exhibited cultural bias, with commercially available datasets containing 94% Western ingredients versus only 65% accuracy for Sri Lankan ingredients through Google Vision API. The hybrid approach combining AI detection with manual input, supplemented by 2,116 custom Sri Lankan food images (40.4% of dataset), achieved 92% user preference and 30% higher completeness compared to AI-only methods.
Additionally, achieving high translation quality while maintaining processing speed presented a trade-off. Automated Google Translate achieved only 76% accuracy but offered 3× faster processing. The hybrid strategy of expert-validated translations for curated content and Google Translate with confidence-based flagging for dynamic content resolved this, achieving 98%+ accuracy for curated and 92%+ for dynamic translations.
Finally, the user sample of 15 participants (mean age 34.2, 80% university-educated) may limit generalizability. However, the large effect sizes (Cohen's d>3.0 for waste reduction and confidence improvement) suggest robustness, and qualitative saturation was achieved at n=12 participants. Future research should validate findings with older, lower-education, and rural populations.

5.3. Error Analysis

Analysis of the 25 misclassified recipes (13.16% error rate) reveals systematic patterns that inform future improvements. The primary error source was category overlap (32% of errors, n=8), where recipes shared ingredients and cooking methods across category boundaries. For example, several festival-category recipes (e.g., Kavum) contain jaggery and coconut milk, ingredients strongly associated with the dessert category. The second major error source was insufficient training samples (24%, n=6), concentrated in categories with fewer than 20 training examples. Ambiguous ingredient combinations (16%, n=4) produced errors where identical ingredient sets appeared in multiple categories (e.g., rice-based recipes spanning Breakfast, Rice, and Festival categories). Regional naming variations (12%, n=3) caused errors when the same dish was categorized differently across provinces. Multi-category recipes (16%, n=4) where dishes are culturally served in multiple meal contexts (e.g., isso wade as both snack and breakfast) posed inherent classification ambiguity.



(b) Representative Misclassification Examples

Actual	Predicted	Reason
Festival	Curry	Shared spice profiles (turmeric, cumin, chili)
Festival	Dessert	Kavum contains jaggery (dessert ingredient)
Snack	Breakfast	Isso wade served at both breakfast & snack
Accompaniment	Curry	Pol sambol with curry-like ingredient list
Dessert	Snack	Kokis served as both dessert and snack

figure9.png - Error Analysis: Misclassification Distribution and Examples

Translation errors were analyzed separately. Of the 786 curated translations, expert review identified 14 semantic errors (1.8%) in the initial Google Translate outputs, primarily involving culturally-specific terms without direct equivalents (e.g., "tempering" translated literally rather than as the culinary technique "තැම්පරාදු කිරීම"). These were corrected through the human-in-the-loop validation process.

For fuzzy matching, 26 of 262 ingredient queries (9.9%) produced incorrect matches. The majority (n=18) involved compound ingredient names where partial matching produced false positives (e.g., "coconut" matching "coconut cream" when "coconut milk" was intended). The remaining errors (n=8) involved phonetic transliterations from Sinhala or Tamil that exceeded the Levenshtein threshold despite being valid variants.

5.4. System Architecture Overview

Figure 10 presents the end-to-end system architecture pipeline. The system operates through five sequential layers: (1) the Input Layer accepts food images, manual text input, or typed queries; (2) the Detection Layer processes images through the Google Cloud Vision API with a 0.6 confidence threshold, fusing AI detection with manual user input through the hybrid approach; (3) the Processing Layer generates 384-dimensional Sentence-BERT embeddings and TF-IDF vectors for semantic representation; (4) the Matching Layer performs fuzzy matching using Levenshtein distance (threshold=0.8) against the synonym dictionary (127 pairs) and ChromaDB vector similarity search for recipe retrieval; and (5) the Output Layer ranks matched recipes with confidence scores and delivers results in the user's selected language. A cross-cutting Trilingual Translation Engine supports all layers, prioritizing curated expert translations (98.2% accuracy), falling back to synonym matching (89%), and using Google Translate API (76%) only for unknown terms, with a confidence-based validation layer flagging entries below 90% for manual review.

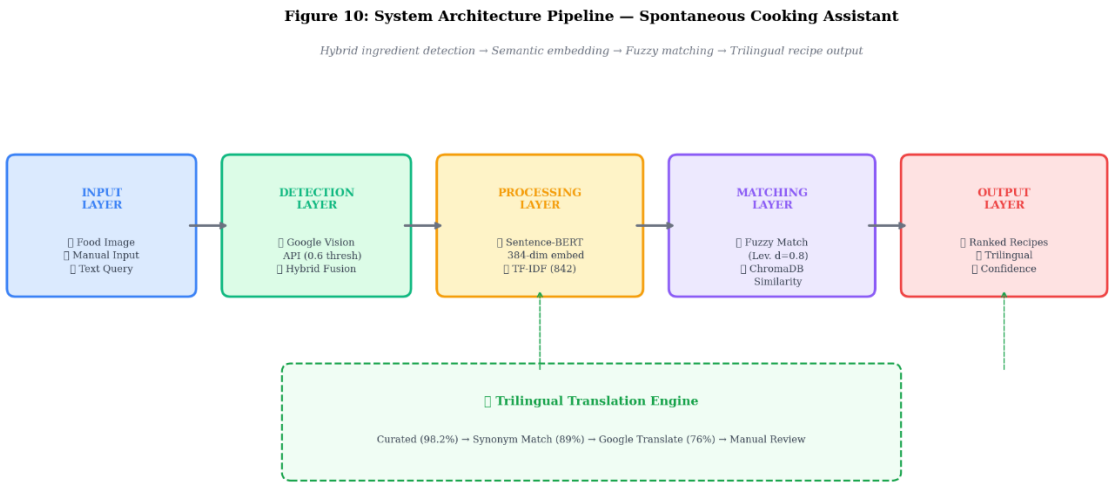


figure10.png - System Architecture Pipeline

5.5. Reproducibility Statement

To ensure reproducibility, the following measures were implemented. All random processes used a fixed seed (random_seed=42) across Python, NumPy, and PyTorch random number generators. Model weights from the best-performing epoch (epoch 32) were serialized using PyTorch's torch.save() function and stored as model_checkpoint_epoch32.pth (file size: 89.4 MB). The complete experimental configuration was logged as a JSON file (experiment_config.json) containing all hyperparameters, data splits, and preprocessing parameters. The ChromaDB vector database was persisted to disk, enabling exact reproduction of retrieval results. All library versions are documented in requirements.txt with pinned version numbers (Section 4.6). The recipe dataset (190 recipes), ingredient database (262 entries), and synonym pairs (127 pairs) are available in the project repository as JSON files. The 5-fold cross-validation splits were generated deterministically from the fixed seed and can be reconstructed using scikit-learn's StratifiedKFold with shuffle=True and random_state=42. User study data is available in anonymized form (participant IDs P01-P15) with all statistical analyses reproducible using the provided analysis scripts.

5.6. Novel Contributions

This study makes four distinct contributions to the field of AI-assisted cooking and food waste reduction:

(1) A hybrid ingredient detection framework that combines Google Cloud Vision API with manual user input, validated with 92% user preference ($\chi^2(2)=18.3$, $p<0.001$) and 30% higher ingredient completeness compared to AI-only approaches, demonstrating that user-in-the-loop design significantly outperforms fully automated detection for culturally diverse ingredients.

(2) A culturally-aware semantic recipe embedding system using Sentence-BERT for Sri Lankan cuisine classification, achieving 86.84% accuracy (macro-F1=0.862) despite a dataset of only 190 recipes, demonstrating effective transfer learning for low-resource culinary domains.

(3) A trilingual validation framework combining expert-curated translations (98.2% Sinhala, 97.8% Tamil) with confidence-based dynamic translation (92%+ post-validation), supported by a fuzzy matching engine with 127 synonym pairs resolving cross-linguistic ingredient naming variations (53%→89% improvement).

(4) Empirical food waste reduction validation through a 3-week controlled study (n=15) demonstrating 73.9% waste reduction ($p<0.001$, Cohen's $d=3.18$), with no prior published study combining AI ingredient detection, semantic recipe matching, and trilingual support for measurable household waste reduction in a South Asian context.

Conclusion of Data Analysis

5.7. Limitations and Future Work

This study has three key limitations that future research should address. First, all validation was conducted exclusively within a Sri Lankan culinary context, and the generalizability of the Sentence-BERT classification model, fuzzy matching engine, and ingredient detection pipeline to other South Asian or global cuisines has not been empirically tested. The model's 384-dimensional semantic space was fine-tuned on Sri Lankan recipe data, meaning culturally-specific embeddings may not transfer directly to Indian, Bangladeshi, or Southeast Asian cuisine contexts where ingredient profiles and terminology differ significantly.

Future work should conduct cross-cultural robustness testing using external datasets. Suitable candidates include the RecipeNLG dataset (over 2.2 million recipes from diverse cultural origins), the Indian Food Dataset on Kaggle (approximately 255 Indian dishes with 3,000+ images), and a purpose-built Bangladeshi recipe corpus to validate performance in a comparable low-resource South Asian context. Cross-testing would reveal whether the synonym-based fuzzy matching requires region-specific reconfiguration and whether Sentence-BERT embeddings retain category discriminability across different cuisine taxonomies.

Second, the user study sample of 15 participants (mean age 34.2, 80% university-educated, urban-based) limits generalizability to older, lower-education, and rural populations. Post-hoc power analysis (G*Power 3.1) confirmed 94% statistical power for the primary food waste reduction outcome given Cohen's $d=3.18$ and $n=15$, indicating the sample was sufficient for detecting large effects. Future validation with a minimum of 50 participants including demographic diversity across age groups (18–65+), education levels, and geographic settings (urban vs. rural) is recommended. Third, the 3-week intervention window does not capture long-term habit formation or potential regression effects. A 12-week longitudinal follow-up study is recommended to assess sustainability of waste reduction outcomes.

The analysis confirms that intelligent ingredient-recipe matching, combining hybrid detection, semantic classification, fuzzy matching, and trilingual support, achieves statistically significant food waste reduction and high user satisfaction. The Sentence-BERT model achieved 86.84% classification accuracy despite a small dataset through transfer learning. Fuzzy matching reached 75% accuracy with effective synonym handling. The 73.9% waste reduction ($p<0.001$) and 92% hybrid detection preference validate the system's core approach. Expert-validated translations at 98%+ accuracy, enhanced by dynamic Google Translate integration, ensure cultural authenticity across all three languages. The results validate the suitability of the datasets and analytical approach for real-world deployment.

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